

Losing Capital Status: Does It Matter for a City's Development?

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Motivation

- Governments periodically redraw administrative maps
 - Here: consolidation of regions
- Different political contexts:
 - Efficiency argument: Vietnam (2025); England (2025-28); Greece (2010); France (2016).
 - EU accession: Moldova (2025-26).
- Consolidation of regions $\implies$ reduction in the number of regional capitals
- Little evidence on the impact of the *loss* of capital status

1975-1998: 49 regions



A map of Poland divided into 16 voivodeships, each labeled with its name in Polish. The voivodeships are: woj. pomorskie (Pomeranian), woj. zachodniopomorskie (West Pomeranian), woj. kujawsko-pomorskie (Kuyavian-Pomeranian), woj. wielkopolskie (Greater Polish), woj. łódzkie (Łódź), woj. śląskie (Silesian), woj. opolskie (Opole), woj. łubuskie (Lubus), woj. dolnośląskie (Lower Silesian), woj. świętokrzyskie (Świętokrzyskie), woj. podkarpackie (Subcarpathian), woj. lubelskie (Lublin), woj. mazowieckie (Masovian), woj. podlaskie (Podlaskian), woj. warmińsko-mazurskie (Warmian-Masurian), and woj. lubelskie (Lublin).

Research question

How did the 1999 loss of capital status impact former capital cities' economic and demographic development?

- Event-study design adjusted for covariates following Callaway and Sant'Anna (2021)

Previous literature...

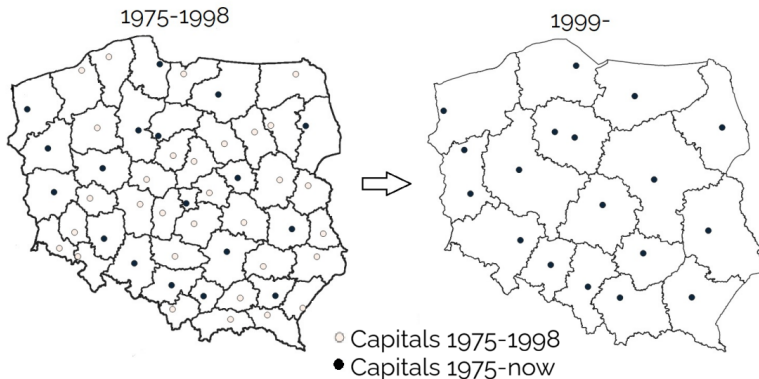
...studies local economic conditions as an outcome of:

- gaining capital status/administrative powers (Becker, Heblich, and Sturm 2021, Faggio, Schlüter, and Berge 2025, Chambru, Henry, and Marx 2022, Bluhm, Lessmann, and Schaudt 2026)
- relocation/changes of public employment (Faggio 2019, Becker, Heblich, and Sturm 2021, Auricchio et al. 2020)

This paper: the first causal analysis of the impact of losing regional capital status

Context

1. Consolidation of regions



2. Creation of second-tier regions (*poviats* - counties)

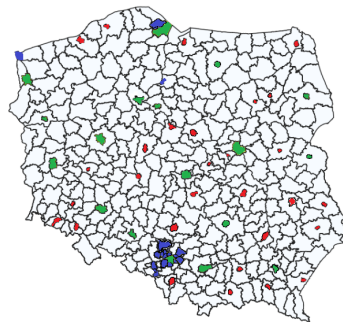


Terminology

“Municipality” / “city” = an **urban commune** (the unit of observation). A **city-county** is a commune that also runs county functions under one government. A **county-seat** is a commune with county and municipal functions separated.

What happened to the biggest Polish cities?

- From former regional capitals to **city-counties** (28)
- From commune seats to **city-counties** (20)
- Remaining **regional capitals** (18)



» More details

Why I exclude remaining capitals from the analysis?

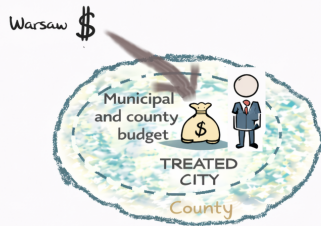
- ① **Remaining a regional capital is a treatment:**
 - Becoming the capital of a much larger region
 - Responsible for the economic development of a region: both legislative and executive functions
 - Disposes of EU funds
- ② **Spillovers from ex-capitals**

Treated municipalities: before and after

Regional capital before 1999:



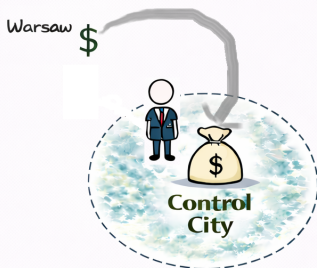
City-county since 1999:



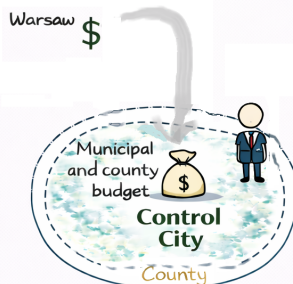
Region > County

Control municipalities: before and after

Municipality before 1999:



City-county since 1999:



Mechanism

Mechanism: scale vs. discretion

Two changes for an ex-capital in 1999

- (−) **Lost** the regional administrative apparatus — offices, public employment, procurement (a loss of **scale**).
- (+) **Gained** city-county status — county budget & tasks folded into the municipality (more local **discretion**).

Public demand = Public budget constraint:

$$D^{g,pre} = \underbrace{M^{pre}}_{\text{local budget}} + \underbrace{\rho R^{pre}}_{\text{regional apparatus}}$$

$$D^{g,post} = \tilde{M}^{post}$$

$$\downarrow D^g \text{ if } : \rho R^{pre} > \tilde{M}^{post} - M^{pre}$$

*Hypothesis: loss of **scale** > gain in **discretion**.*

Predictions: who is exposed to public demand?

Public spending splits into two uses:

$$D^g = \underbrace{S^g}_{\text{services}} + \underbrace{K^g}_{\text{capital / infrastructure}}$$

Private output of sector s : $Y_s^p = \lambda_{sS} S^g + \lambda_{sK} K^g + X_s$

Sector	λ_{sS}	λ_{sK}
Construction	low	high
Real estate	mod.	high
Other services, retail	high	low
Manufacturing	low	low

Testable predictions:

- P1. ↓ **capital/infrastructure** sectors (construction, real estate)
- P2. ↓ **services** sectors (local services, admin support)
- P3. transfers help only if they restore the lost **services / capital** mix

Data and identification strategy

Data

Municipality-level data (1995–2003)

- 48 urban municipalities:
 - 28 former regional capitals (treated)
 - 20 other city-counties (controls)
- Ongoing digitization of administrative and fiscal data

Outcomes

- Municipal finance
- Labor market
- Firms
- Demographics

Policy shock: administrative reform implemented in 1999

►► Summary statistics

Identification strategy

Treated

- Former regional capitals
- Lost capital status in 1999
- Became city-counties

Controls

- Never regional capitals
- Also became city-counties in 1999

Key idea: all municipalities are affected by the reform
→ identification comes from **differential exposure**

Estimand

$$ATT = E[Y(1) - Y(0) \mid \text{former capital}]$$

Identification: assumptions

- No anticipation [» Anecdotal evidence](#)
- Parallel trends $\implies$ “honest-DiD”: Rambachan and Roth 2023 [» Intuition](#)
- SUTVA $\implies$ interact treatment with distance to the nearest remaining capital [» Discussion and test description](#)
- **The city-county upgrade affected both groups equally**
- Other issues:
 - Spatial congestion of controls $\implies$ estimation excl. Silesia region
 - Many outcomes $\implies$ multiple testing [» Link](#)

Robustness: heterogeneous city-county effects

Concern:

- The city-county reform may affect former capitals and other cities differently

Decomposition (main comparison):

$$\beta^{main} = \beta^{loss} + \Delta$$

Interpretation:

- β^{loss} : effect of losing capital status
- $\Delta = \beta^{city-county, tr} - \beta^{city-county, control}$

Key challenge:

- Both components of Δ are not directly observed in the main comparison

Robustness: heterogeneous city-county effects

Bounding strategy:

- Estimate upgrade effect in a **separate** comparison:

$$\beta^{upgrade} \quad (\text{city-counties vs county seats})$$

- Use it as a benchmark for heterogeneity:

$$|\Delta| \leq |\beta^{upgrade}|$$

Implied bounds:

$$\beta^{loss} \in \left[\beta^{main} - |\beta^{upgrade}|, \beta^{main} + |\beta^{upgrade}| \right]$$

Interpretation:

- Allows the effect of becoming a city-county to be different between treated and control municipalities

Baseline event study design

$$Y_{it} = \alpha_i + \alpha_t + \sum_{k \neq -1} \beta_k \cdot 1\{\text{event time} = k\} + \varepsilon_{it}$$

where:

- α_i : municipality fixed effects; α_t : year fixed effects
- ε_{it} clustered at 50km radius

Event study with covariate adjustment (Callaway and Sant'Anna 2021)

$$ATT_t = \mathbb{E}[Y_t(1) - Y_t(0) \mid D = 1]$$

Implementation:

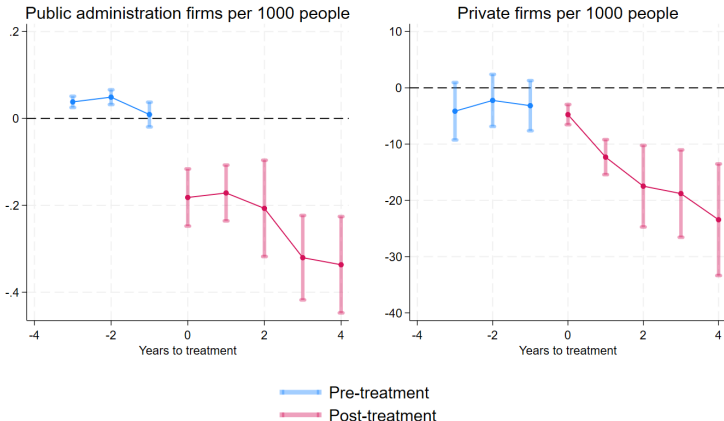
$$Y_{it} = \alpha_i + \alpha_t + \sum_{k=-K}^{-2} \delta_k^{anticip} \cdot D_{it}^k + \sum_{k=0}^L \beta_k \cdot D_{it}^k + \epsilon_{it},$$

- Estimator for β_k : doubly-robust

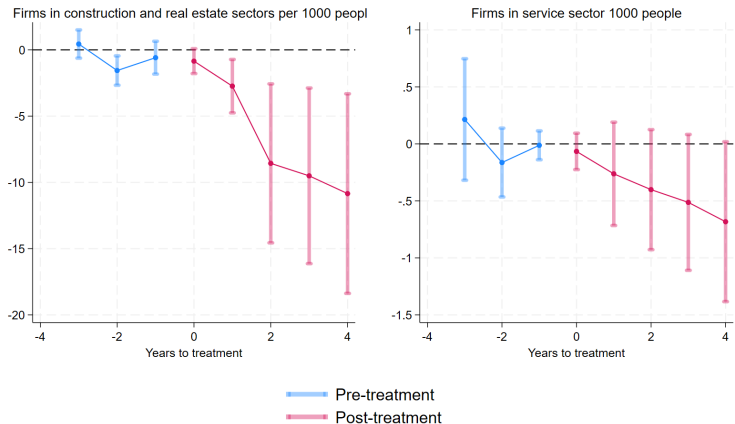
► Estimator in details

Results

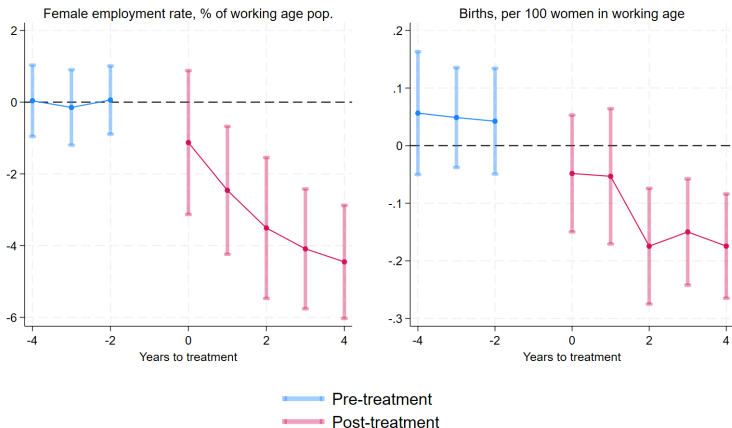
Result 1: Firms in public and private sector



Result 2: Significant decline in construction, but not services



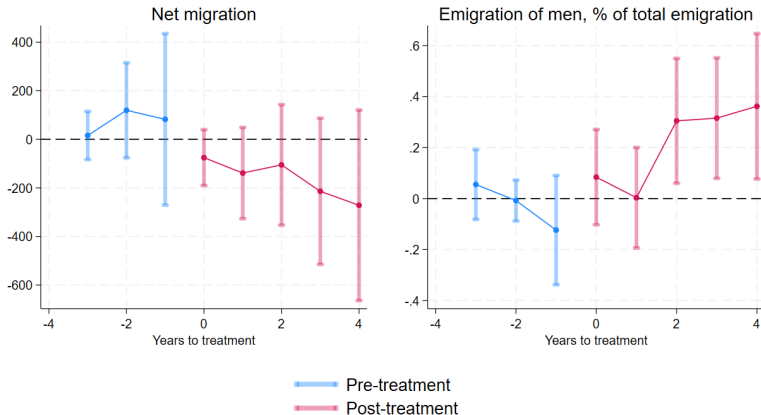
Result 3: Women significantly affected



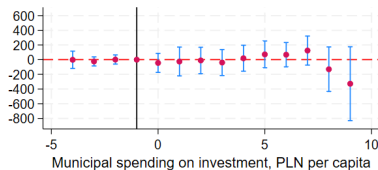
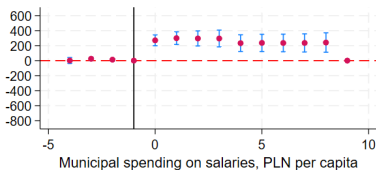
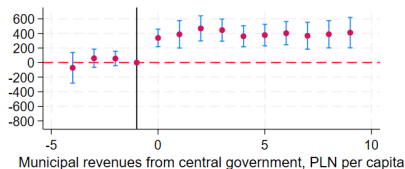
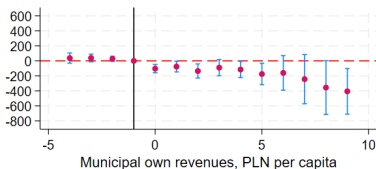
Female employment: $\downarrow$ 3 p.p. $\approx$ 1,000 jobs per city

Births: $\approx$ 5% of the 1998 $\rightarrow$ 2003 TFR decline.

Result 4: Imprecise effect in migration



Channel: Municipal finance



Own revenue ↓: a weaker local tax base.

Transfers $\uparrow \approx \$77/\text{capita}/\text{yr}$ — but spent on **salaries, not investment**.

Conclusion



Losing capital status **matters**, even with greater fiscal autonomy and compensation: lost scale > gained discretion.



Public sector firms — the direct effect (robust).



Construction and real estate fall, but not services.



Women significantly affected: female employment & fertility decline.



Transfers spent on **salaries, not investment**.



Migration effect imprecise; men more likely to leave ex-capitals.

Policy implication

The 1999 administrative reform was needed to prepare Poland for EU accession.

What should the government have done differently?

- Support former public employees in the transition
- Help the private sector in a new landscape with less demand from public administration



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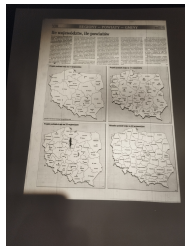
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"We learned about the reform process around half a year before its introduction, but it was too late. Nothing had been prepared for this administrative reform apart from the legislation in Warsaw."

— Former employee, Regional Office of Łomża (ex-capital)



Political cartoon, 1998:
"I see 9 regions, 8 duchies, and 2
republics!"

Four competing reform proposals
circulated in the press in mid-1998
(12–27 regions)

Municipal finance

	Ex-capitals		City-counties	
	1995–98	1999–03	1995–98	1999–03
<i>(Real PLN per capita, 1994=100)</i>				
Own revenues	177.68 (39.13)	331.79 (77.21)	197.28 (77.99)	401.0 (116.0)
Central transfers	184.24 (39.13)	530.62 (77.21)	213.40 (77.99)	383.21 (116.0)
Investment	109.11 (48.17)	109.38 (52.59)	112.30 (79.91)	117.83 (138.0)
Salaries	116.41 (54.0)	360.0 (74.5)	160.65 (43.18)	275.54 (47.82)

Table 1: Summary statistics: mean and standard deviation

$$\hat{\beta}_k \approx \underbrace{\tau_k}_{\text{effect (=0 pre)}} + \underbrace{\delta_k}_{\text{PT violation}} \Rightarrow \text{pre-period } \hat{\beta}_k \text{ measure } \delta.$$

Relative magnitudes $\Delta^{RM}(\bar{M})$

Harsh when: pre-trends are **noisy** (anchors to noise).

Smoothness $\Delta^{SD}(M)$

Harsh when: pre-trends are **steep & linear**.

▶▶ Back

Event study with covariate adjustment (Callaway and Sant'Anna 2021)

$$Y_{it} = \alpha_i + \alpha_t + \sum_{k=-K}^{-2} \delta_k^{anticip} \cdot D_{it}^k + \sum_{k=0}^L \beta_k \cdot D_{it}^k + \epsilon_{it},$$

Estimator for β_k :

$$\hat{\beta}_k = \widehat{\text{ATT}}_k^{DR} = \frac{1}{N_{tr}} \sum_{treated} [Y_{it} - \hat{\mu}_0(X_i, t)] - \frac{1}{N_0} \sum_{control} \hat{w}_i(X_i) \cdot [Y_{it} - \hat{\mu}_0(X_i, t)]$$

where:

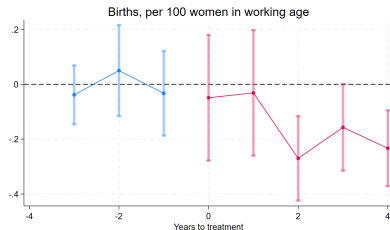
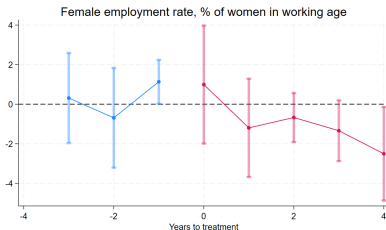
- $\hat{\mu}_0(X_i, t)$ is the predicted untreated outcome from a regression of Y_{it} on covariates X_i using only never-treated units
- $\hat{w}_i(X_i)$ are inverse probability weights estimated from a propensity score model for treatment assignment.

Sensitivity to heterogeneous city-county effects

Outcome	β^{main}	$\beta^{upgrade} \geq \Delta$	Lower bound	Upper bound
Female employment	-3.13 (0.84)	1.64 (1.92)	-4.76	-1.49
Births	-0.12 (0.04)	0.04 (0.04)	-0.16	-0.08

Bounds use the city-county vs county-seat comparison to allow for heterogeneous upgrade effects.

Excluded control cities from Silesia region



►► Back

Female employment rate: Effect of the distance to remaining capital

d_i = km to nearest retained capital. Treated are far ($\bar{d}=77$), controls close ($\bar{d}=22$) — structural: consolidation keeps capitals spread out.

Levels need not balance — DiD needs parallel trends

$$y_{it} = \alpha_i + \delta_t + \beta (T_i \times \text{post}_t) + \gamma (\text{post}_t \times d_i) + \theta (T_i \times \text{post}_t \times d_i) + \varepsilon_{it}$$

	coef.	p
β (loss effect at mean d)	-2.64	0.03
γ (post $\times$ distance)	-0.03	0.16
θ (loss $\times$ distance)	0.04	0.22

Births per 100 women in working age

Effect of the distance to remaining capital

d_i = km to nearest retained capital. Treated are far ($\bar{d}=77$), controls close ($\bar{d}=22$) — structural: consolidation keeps capitals spread out.

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	coef.	p
β (loss effect at mean d)	-0.06	0.41
γ (post $\times$ distance)	-0.002	0.21
θ (loss $\times$ distance)	0.001	0.59

» Back

Sensitivity to violations of (conditional) parallel trends (Rambachan and Roth, 2023)

Restriction	Relative magnitudes $\bar{M}$		Smoothness M	
	Lower	Upper	Lower	Upper
Female employment rate				
Original	-4.77	-1.48	-4.77	-1.48
0.00	-4.75	-1.50	-4.42	-1.64
0.50	-6.70	-0.10	-8.23	2.10
1.00	-9.06	1.63	-11.86	5.83
1.50	-11.47	3.45	-15.36	9.45
2.00	-13.97	5.87	-18.85	12.96

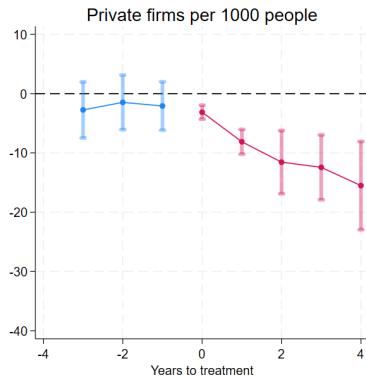
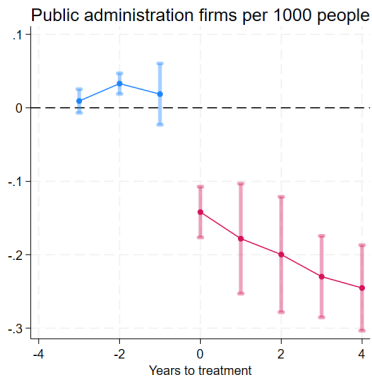
Notes: Robust 95% confidence sets for the average post-treatment effect under restrictions on parallel-trends violations, on the covariate-adjusted Callaway–Sant’Anna event study (doubly robust, universal base period). is the unrestricted CI; $\bar{M}$ scales post-treatment violations by the largest pre-treatment violation; M bounds the change in slope. SEs clustered at 50 km.

Sensitivity to violations of (conditional) parallel trends (Rambachan and Roth, 2023)

Restriction	Relative magnitudes $\bar{M}$		Smoothness M	
	Lower	Upper	Lower	Upper
Births per 100 women in working age				
Original	-0.20	-0.04	-0.20	-0.04
0.00	-0.20	-0.04	-0.21	0.04
0.50	-0.29	0.11	-3.77	3.78
1.00	-0.46	0.28	-7.27	7.28
1.50	-0.63	0.46	-10.77	10.78
2.00	-0.81	0.64	-14.27	14.28

Notes: Robust 95% confidence sets for the average post-treatment effect under restrictions on parallel-trends violations, on the covariate-adjusted Callaway–Sant’Anna event study (doubly robust, universal base period). is the unrestricted CI; $\bar{M}$ scales post-treatment violations by the largest pre-treatment violation; M bounds the change in slope. SEs clustered at 50 km.

Excluded control cities from Silesia region



—●— Pre-treatment
—●— Post-treatment

Public administration firms per 1000 inhabitants: Effect of the distance to remaining capital

d_i = km to nearest retained capital. Treated are far ($\bar{d}=77$), controls close ($\bar{d}=22$) — structural: consolidation keeps capitals spread out.

Levels need not balance — DiD needs parallel trends

$$y_{it} = \alpha_i + \delta_t + \beta (T_i \times \text{post}_t) + \gamma (\text{post}_t \times d_i) + \theta (T_i \times \text{post}_t \times d_i) + \varepsilon_{it}$$

	coef.	<i>p</i>
β (loss effect at mean d)	−0.17	0.00
γ (post × distance)	0.001	0.16
θ (loss × distance)	−0.0007	0.48

Private firms per 1000 inhabitants: Effect of the distance to remaining capital

d_i = km to nearest retained capital. Treated are far ($\bar{d}=77$), controls close ($\bar{d}=22$) — structural: consolidation keeps capitals spread out.

Levels need not balance — DiD needs parallel trends

$$y_{it} = \alpha_i + \delta_t + \beta (T_i \times \text{post}_t) + \gamma (\text{post}_t \times d_i) + \theta (T_i \times \text{post}_t \times d_i) + \varepsilon_{it}$$

	coef.	p
β (loss effect at mean d)	-7.21	0.07
γ (post $\times$ distance)	0.21	0.05
θ (loss $\times$ distance)	-0.10	0.47

» Back

Sensitivity to violations of (conditional) parallel trends (Rambachan and Roth, 2023)

Restriction	Relative magnitudes $\bar{M}$		Smoothness M	
	Lower	Upper	Lower	Upper
Private firms				
Original	-11.10	-0.59	-11.10	-0.59
0.00	-11.05	-0.64	-9.29	0.40
0.50	-19.47	9.56	-12.46	2.72
1.00	-32.04	22.44	-15.96	6.22
1.50	-45.05	35.49	-19.46	9.72
2.00	-58.26	48.57	-22.96	13.22

Notes: Robust 95% confidence sets for the average post-treatment effect under restrictions on parallel-trends violations, on the covariate-adjusted Callaway–Sant’Anna event study (doubly robust, universal base period). is the unrestricted CI; $\bar{M}$ scales post-treatment violations by the largest pre-treatment violation; M bounds the change in slope ($M=0$ permits a linear pre-trend). SEs clustered at 50 km.

Sensitivity to violations of (conditional) parallel trends (Rambachan and Roth, 2023)

Restriction	Relative magnitudes $\bar{M}$		Smoothness M	
	Lower	Upper	Lower	Upper
Public administration firms				
Original	-0.27	-0.13	-0.27	-0.13
0.00	-0.27	-0.13	-0.32	-0.17
0.50	-0.33	-0.09	-3.91	3.40
1.00	-0.41	-0.03	-7.41	6.90
1.50	-0.49	0.04	-10.91	10.40
2.00	-0.57	0.11	-14.41	13.90

Notes: Robust 95% confidence sets for the average post-treatment effect under restrictions on parallel-trends violations, on the covariate-adjusted Callaway–Sant’Anna event study (doubly robust, universal base period). is the unrestricted CI; $\bar{M}$ scales post-treatment violations by the largest pre-treatment violation; M bounds the change in slope. SEs clustered at 50 km.

What happened to Polish cities?

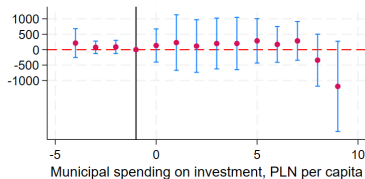
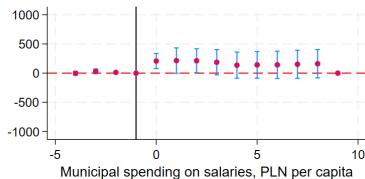
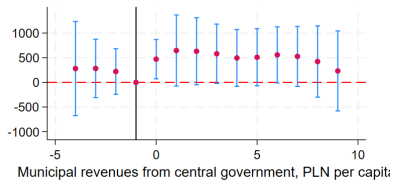
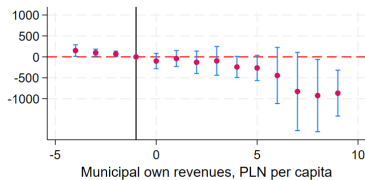
Before 1999 a city could be:	Since 1999 a city can be:	#
Regional capital + Commune seat	Regional capital + City-county + Commune seat	18
	City-county + Commune seat	28
	County seat + Commune seat	3
Commune seat	City-county + Commune seat	20
	County seat + Commune seat	314
	Commune seat	2774

» Back

Treated municipalities: before and after

Regional capital before 1999:	City-county since 1999:
Seat of centrally-appointed regional governors	Seat of locally-appointed mayors of city and county
Regional Office	City-county Office with some of the Regional Office staff
Revenue from central gov., limited local discretion in spending	Revenue split between locally collected taxes and central transfers

Excluded control cities from Silesia region



 Point Estimate
 95% CI

Municipal own revenues per capita: Effect of the distance to remaining capital

d_i = km to nearest retained capital. Treated are far ($\bar{d}=77$), controls close ($\bar{d}=22$) — structural: consolidation keeps capitals spread out.

Levels need not balance — DiD needs parallel trends

$$y_{it} = \alpha_i + \delta_t + \beta (T_i \times \text{post}_t) + \gamma (\text{post}_t \times d_i) + \theta (T_i \times \text{post}_t \times d_i) + \varepsilon_{it}$$

	coef.	p
β (loss effect at mean d)	-111.39	0.32
γ (post $\times$ distance)	-3.60	0.36
θ (loss $\times$ distance)	4.35	0.30

Municipal spending on salaries per capita: Effect of the distance to remaining capital

d_i = km to nearest retained capital. Treated are far ($\bar{d}=77$), controls close ($\bar{d}=22$) — structural: consolidation keeps capitals spread out.

Levels need not balance — DiD needs parallel trends

$$y_{it} = \alpha_i + \delta_t + \beta (T_i \times \text{post}_t) + \gamma (\text{post}_t \times d_i) + \theta (T_i \times \text{post}_t \times d_i) + \varepsilon_{it}$$

	coef.	<i>p</i>
β (loss effect at mean d)	213.39	0.05
γ (post $\times$ distance)	2.22	0.31
θ (loss $\times$ distance)	-3.73	0.16

Sensitivity to violations of unconditional parallel trends (Rambachan and Roth, 2023)

Table 4: Honest DiD: Own revenue (baseline event study)

	Relative magnitudes $\bar{M}$		Smoothness M	
M	Lower	Upper	Lower	Upper
Original	-335.82	-36.86	-335.82	-36.86
0.00	-334.44	-38.24	-232.58	-33.52
0.50	-454.00	10.80	-234.74	-28.08
1.00	-587.54	93.99	-240.48	-14.04
1.50	-728.49	222.47	-248.64	3.46
2.00	-872.76	364.62	-258.04	21.30

Notes: Honest-DiD sensitivity (Rambachan and Roth 2023) for the average post-treatment effect, baseline TWFE event study (eq. 18), reference $k = -1$. $\bar{M}$ = relative magnitudes; M = smoothness ($M=0$ permits a linear pre-trend). SEs clustered at 50 km.

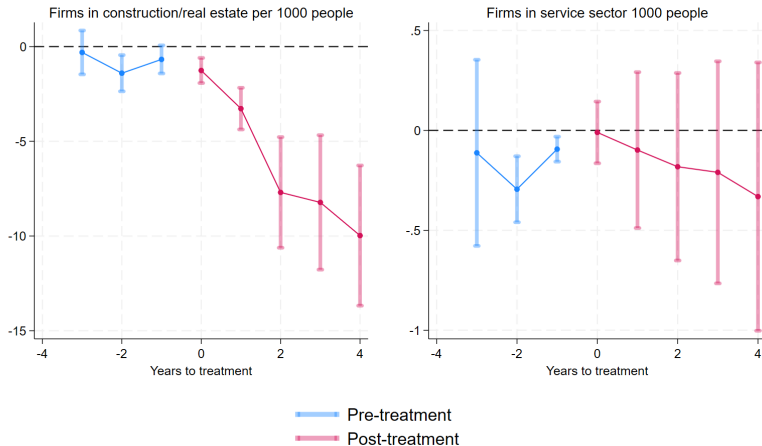
Sensitivity to violations of unconditional parallel trends (Rambachan and Roth, 2023)

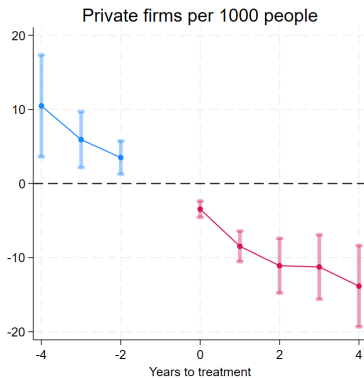
Table 5: Honest DiD: Central transfers (baseline event study)

	Relative magnitudes $\bar{M}$		Smoothness M	
M	Lower	Upper	Lower	Upper
Original	251.69	538.51	251.69	538.51
0.00	255.94	534.26	136.37	652.26
0.50	-260.78	934.05	134.46	653.60
1.00	-815.45	1487.44	128.80	657.46
1.50	-1400.00	2040.61	119.76	663.63
2.00	-1900.00	2600.46	107.93	671.76

Notes: Honest-DiD sensitivity (Rambachan and Roth 2023) for the average post-treatment effect, baseline TWFE event study (eq. 18), reference $k = -1$. $\bar{M}$ = relative magnitudes; M = smoothness ($M=0$ permits a linear pre-trend). SEs clustered at 50 km.

Excluded control cities from Silesia region





▶▶ Back

Multiple testing: Benjamini–Hochberg

Table 8: Multiple-testing adjustment by outcome family

Outcome	ATT	p	q (FDR)
<i>Panel A. Primary outcomes (P1–P3), $m = 6$</i>			
Public administration firms	-0.201	<0.001	<0.001
Construction / real estate firms	-6.500	0.005	0.008
Services firms	-0.385	0.115	0.139
Female employment	-3.126	<0.001	<0.001
Male employment	-1.282	0.248	0.248
Municipal own revenue	-109.843	0.004	0.007
<i>Panel B. Fiscal channel, $m = 3$</i>			
Central government transfers	400.460	<0.001	<0.001
Spending on salaries	285.406	<0.001	<0.001
Spending on investment	-16.428	0.826	0.826

Notes: ATTs from Callaway–Sant’Anna (universal base period); SEs clustered at 50 km; ATT in each outcome’s own units. q = Benjamini–Hochberg FDR q -values, computed within each pre-specified family. Panel A includes the null results (services, male employment); the investment null in Panel B is itself the finding (“salaries, not investment”). Sharpened two-stage q -values (Anderson 2008) are uniformly smaller.

Figure 1 consists of two side-by-side scatter plots with error bars, showing employment trends from 1995 to 2010. The left plot is titled 'Employment of men, % of men in working age' and the right plot is titled 'Employment of women, % of women in working age'. Both plots share a common x-axis representing years from 1995 to 2010, with major ticks every five years. A vertical dashed line is drawn at the year 1999 in both plots. The legend at the bottom indicates two groups: 'Control: county seats' represented by blue dots and 'Treated: ex-capital+city counties' represented by red dots. Each data point includes a vertical error bar. In the men's plot, the y-axis ranges from 20 to 70. In the women's plot, the y-axis ranges from 20 to 60. In both plots, the treated group (red) shows a notable increase in employment levels after 1999 compared to the control group (blue).